

Site-Specific Probing Beams for Full-Duplex Millimeter Wave Systems

Abraham Canafe and Ian P. Roberts

Abstract

This work develops a novel algorithm that simultaneously performs beam alignment and self-interference channel estimation algorithm in a full-duplex millimeter-wave (mmWave) setting. Our proposed method leverages a site-specific probing codebook that not only captures important characteristics about the propagation environment, but also can be harnessed by a neural network (NN) to generate accurate estimates of the self-interference (SI) channel. Our experimental results show that the use of probing and sensing codebooks can feasibly be used to simultaneously estimate self-interference coupling during the beam alignment process.

I. INTRODUCTION

Recent efforts dedicated to equipping millimeter-wave (mmWave) communication systems with full-duplex capabilities have been spearheaded by beamforming techniques that can spatially mitigate the consequent self-interference (SI) coupling that manifests. Whereas a half-duplex mmWave base station (BS) can straightforwardly rely on conventional beamforming codebooks [1] to provide coverage for a region, a full-duplex BS can employ carefully designed codebooks that can simultaneously service a site and ameliorate the prohibitive effects of self-interference [2], [3]. Works such as STEER [4] and STEER+ [5] have sought to address the challenge of mitigating self-interference by leveraging a joint selection of transmit and receive beams designed to ensure high gain in the desired steering directions while maintaining low self-interference coupling. In another work [6], LONESTAR codebooks, were designed to minimize self-interference coupling for every transmit-receive beam pair within the codebooks. Compared to STEER, which exploits measurements of candidate beam pairs during the beamforming process and thus does not require direct knowledge of the self-interference channel, LONESTAR conversely relies on estimates of the SI coupling in order to construct its codebooks.

Indeed, an effective cancellation of self-interference coupling in full-duplex systems seems to require reliable SI channel estimates [3], [6]. Least squares (LS) channel estimation strategies have been employed in studies of full-duplex systems such as [7], which analyzed the achievable sum rate for bidirectional communication between a pair of transceivers, and [8], which examined the achievable rates of a multipair two-pair amplify-and-forward (AF) relaying system equipped with a low-pilot overhead scheme. The work carried out in [9] utilized minimum mean square error (MMSE) to estimate self-interference coupling, as well as the user equipment channels (UE), to study the ergodic achievable sum rates for cooperative and non-cooperative multicell scenarios. More recently, [10] addressed the problem of SI coupling by employing simple neural networks (NNs) to estimate both the self-interference channel at the base station (BS) and the user equipment channels.

Still, the task of efficiently and reliably estimating the self-interference channels is a highly difficult one and remains unsolved. This difficulty is also accompanied by the challenge of leveraging beam sweeping and well-designed codebooks to accurately conduct beam alignment in the current 5G framework [11], [12]. Beam alignment procedures that sweep over traditional codebooks, which are designed for a narrow frequency range, do not perform well during multi-band operations [12]. Beam sweeping approaches involving the use of hierarchical codebooks designed to reduce beam sweeping complexity, such as those used in [13]–[16], can suffer from hierarchical beam search errors that stem from measurement noise, imperfect wide-beam patterns, and multi-path channels [11], [12].

A NN-based Methodology for Site-Specific Beam Alignment and Self-Interference Estimation. The need for a strategic approach to beam alignment and the estimation of the self-interference channel in full-duplex mmWave systems has motivated us to create a novel methodology that confronts both challenges. Inspired by [11] and [17], we utilize a trainable site-specific probing codebook at the BS transmitter that can measure site-specific propagation characteristics in order to optimize beam selection for downlink communications. We propose an algorithm that critically demonstrates the feasibility of utilizing probing and sensing codebooks in order to efficiently estimate the self-interference channel while jointly optimizing the beam alignment process. This estimate can then be used for SI cancellation schemes such as LONESTAR or even for sensing.

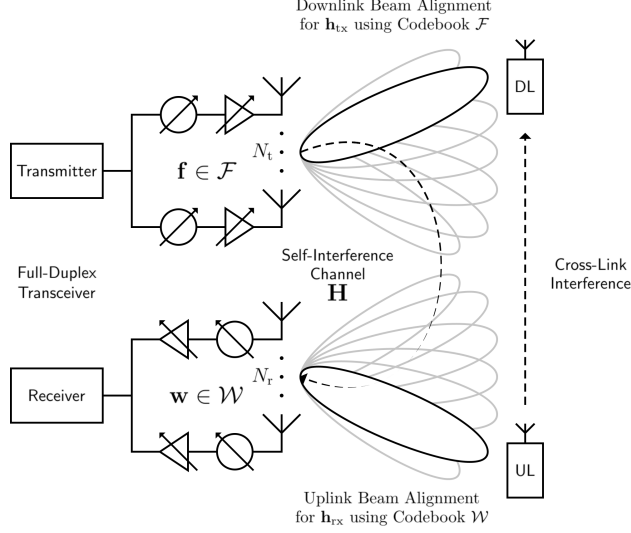


Fig. 1. A full-duplex mmWave system carries out in-band communications with uplink and downlink users. The probing and sensing codebooks, $\mathcal{F}$ and $\mathcal{W}$, are used to estimate the self-interference channel $\mathbf{H}$. The codebook $\mathcal{F}$ is simultaneously used to capture site-specific measurements in order to enhance beam alignment for the downlink channel, $\mathbf{h}_{tx}$. In this paper, we design $\mathcal{F}$ to be a trainable complex module, and we treat $\mathcal{W}$ as a discrete Fourier transform (DFT) codebook. Adapted from [6].

II. SYSTEM MODEL

We consider a full-duplex mmWave system BS which utilizes transmit antenna arrays and receive antenna arrays designed to separately serve downlink and uplink users as illustrated in Fig. 1. We use N_t and N_r to represent the number of transmit and receive antennas at the BS. We denote the transmit and receive array response vectors, in some direction θ , as $\mathbf{a}_{tx}(\theta) \in \mathbb{C}^{N_t \times 1}$ and $\mathbf{a}_{rx}(\theta) \in \mathbb{C}^{N_r \times 1}$, observing that the array response vectors for any θ can be computed based on the BS array geometries [18], [19]. We also use the convention $\|\mathbf{a}_{tx}(\theta)\|_2^2 = 1$ and $\|\mathbf{a}_{rx}(\theta)\|_2^2 = 1$, and we set forth two simplifying assumptions: that the user devices are single-antenna devices and that the BS antenna arrays are uniform linear arrays (ULAs). We further assume that the transmit array and receive array at the BS perform analog or hybrid beamforming. We denote $\mathbf{v}_{tx} \in \mathbb{C}^{N_t \times 1}$ as the vector of beamforming weights used at the BS transmitter during downlink communication with the user equipment, and we denote $\mathbf{v}_{rx} \in \mathbb{C}^{N_r \times 1}$ as the beamforming vector used at the BS receiver during uplink. We apply a per-antenna power constraint at the transmitter and receiver so that the beamforming weights do not exceed unit magnitude, leading to

$$|[\mathbf{v}_{tx}]_i| = \frac{1}{\sqrt{N_t}}, \quad \forall i = 1, \dots, N_t \quad (1)$$

$$|[\mathbf{v}_{\text{rx}}]_j| = \frac{1}{\sqrt{N_r}}, \quad \forall j = 1, \dots, N_r. \quad (2)$$

Satisfying the constraints (1) and (2) requires that only phase shifters are used during the beamforming process.

Let us now consider the channel through which the full-duplex BS communicates with the user equipment. We denote $\mathbf{h}_{\text{tx}}^{\text{H}} \in \mathbb{C}^{1 \times N_t}$ as the downlink channel from the BS to the user equipment, and we let $\mathbf{h}_{\text{rx}} \in \mathbb{C}^{N_r \times 1}$ denote the uplink channel from the user equipment to the BS. When the BS transmits a symbol $s \in \mathbb{C}$ in the downlink under the average power constraint $\mathbb{E}[|s|^2] = 1$ using the beamforming vector $\mathbf{v}_{\text{tx}} \in \mathbb{C}^{N_t \times 1}$, the received signal at the user can be modeled as

$$y = \sqrt{P_{\text{tx}}^{\text{BS}}} \mathbf{h}_{\text{tx}}^{\text{H}} \mathbf{v}_{\text{tx}} s + n_{\text{UE}}, \quad (3)$$

where $P_{\text{tx}}^{\text{BS}}$ is the transmit power and $n_{\text{UE}} \sim \mathcal{N}_{\mathbb{C}}(0, \sigma_{\text{UE}}^2)$ is complex Gaussian noise at the user device. Thus, the SNR of a beamformed signal at a user device can be formulated as

$$\text{SNR}_{\text{UE}} = \frac{P_{\text{tx}}^{\text{BS}} |\mathbf{h}_{\text{tx}}^{\text{H}} \mathbf{v}_{\text{tx}}|^2}{\sigma_{\text{UE}}^2}. \quad (4)$$

In this paper, we abstract out complex noise n_{UE} and assume that $\sigma_{\text{UE}}^2 = -\infty$ dB.

Since our full-duplex system performs simultaneous, in-band transmission and reception, a self-interference channel $\mathbf{H} \in \mathbb{C}^{N_r \times N_t}$ exists between the transmit array and receive array at the BS. Although there does not yet exist a measurement-backed model for $\mathbf{H}$ at mmWave frequencies [2], [3], [6], we nevertheless assume that $\mathbf{H}$ is defined by the following Rician channel model with Rice factor $\kappa \in \mathbb{R}^+$

$$\mathbf{H} = \sqrt{\frac{\kappa}{1 + \kappa}} \mathbf{H}_{\text{nf}} + \sqrt{\frac{1}{1 + \kappa}} \mathbf{H}_{\text{ff}}, \quad (5)$$

where $\mathbf{H}_{\text{nf}} \in \mathbb{C}^{N_r \times N_t}$ and $\mathbf{H}_{\text{ff}} \in \mathbb{C}^{N_r \times N_t}$ denote the near-field and far-field components of $\mathbf{H}$ [18]. We characterize the near-field component $\mathbf{H}_{\text{nf}}$ of the self-interference channel $\mathbf{H}$ using the spherical-wave channel model [20] defined by

$$[\mathbf{H}_{\text{nf}}]_{v,u} = \frac{\rho}{r_{u,v}} \exp\left(-j2\pi \frac{r_{u,v}}{\lambda}\right), \quad (6)$$

where $r_{u,v}$ measures the distance between the u -th transmit antenna and the v -th receive antenna, λ represents the carrier wavelength, and ρ serves as the normalizing factor that enforces $\|\mathbf{H}_{\text{nf}}\|_{\text{F}}^2 = N_t \cdot N_r$. We define the far-field component $\mathbf{H}_{\text{ff}}$ as a ray-based narrowband block-fading

mmWave channel model parameterized by complex gains $\beta_m \sim \mathcal{N}_{\mathbb{C}}(0, 1)$, angles of arrival (AoA) θ_m^{AoA} , angles of departure (AoD) θ_m^{AoD} , and M_p paths [21]:

$$\mathbf{H}_{\text{ff}} = \frac{1}{\sqrt{M_p}} \sum_{m=1}^{M_p} \beta_m \mathbf{a}_{\text{rx}}(\theta_m^{\text{AoA}}) \mathbf{a}_{\text{tx}}(\theta_m^{\text{AoD}})^H. \quad (7)$$

III. PROBLEM FORMULATION

Using our system model, we now further formulate our proposed algorithm. We let the transmit array at the BS employ a codebook $\mathbf{V}_{\text{tx}} \in \mathbb{C}^{N_t \times N_{\mathbf{V}_{\text{tx}}}}$ whose columns represent candidate narrow analog beamforming weights for a beam. We assume that $N_{\mathbf{V}_{\text{tx}}}$ is sufficiently large so that the narrowbeam codebook $\mathbf{V}_{\text{tx}}$ can be used to sweep across the entire angular space, and we impose the following constraint on $\mathbf{V}_{\text{tx}}$, following that of (1).

$$|[\mathbf{V}_{\text{tx}}]_{i,j}| = \frac{1}{\sqrt{N_t}}, \quad \forall i = 1, \dots, N_t, \quad \forall j = 1, \dots, N_{\mathbf{V}_{\text{tx}}} \quad (8)$$

The optimal narrow beam weights used to transmit a signal from the BS to a user device satisfies the formulation

$$k_{\mathbf{V}_{\text{tx}}}^* = \underset{k \in \{1, 2, \dots, N_{\mathbf{V}_{\text{tx}}}\}}{\text{argmax}} \left(\frac{|\mathbf{h}_{\text{tx}}^H[\mathbf{V}_{\text{tx}}]_{:,k}|^2 P_{\text{tx}}^{\text{BS}}}{\sigma_{\text{UE}}^2} \right) = \underset{k \in \{1, 2, \dots, N_{\mathbf{V}_{\text{tx}}}\}}{\text{argmax}} \left(|\mathbf{h}_{\text{tx}}^H[\mathbf{V}_{\text{tx}}]_{:,k}|^2 \right), \quad (9)$$

where $k_{\mathbf{V}_{\text{tx}}}^*$ indexes the column associated with the optimal analog beam in codebook $\mathbf{V}_{\text{tx}}$. We assume $\mathbf{h}_{\text{tx}}$ follows a ray-based narrowband block-fading model with N_p paths [21]

$$\mathbf{h}_{\text{tx}} = \sum_{\ell=1}^{N_p} \alpha_{\ell} \mathbf{a}_{\text{tx}}(\theta_{\ell}), \quad (10)$$

where $\alpha_{\ell} \in \mathbb{C}$ is the complex gain of the ℓ -th path.

In the first stage of our proposed algorithm, the BS sweeps across a set of site-specific probing beams from the columns of the probing codebook $\mathbf{F} \in \mathbb{C}^{N_t \times N_{\mathbf{F}}}$, and the user devices subsequently measure and report back to the BS the received power of the probing signals. The received probing beam at the user device is denoted by

$$y_j = \sqrt{P_{\text{tx}}^{\text{BS}}} \mathbf{h}_{\text{tx}}^H[\mathbf{F}]_{:,j} s + n_{\text{UE}}, \quad j = 1, \dots, N_{\mathbf{F}}, \quad (11)$$

and the reported beam sweeping results across $\mathbf{F}$ for each user device is defined by

$$\mathbf{x} = \begin{bmatrix} |y_1|^2 & \dots & |y_{N_{\mathbf{F}}}|^2 \end{bmatrix}^T. \quad (12)$$

The transmitter at the BS simultaneously uses sensing beams from $\mathbf{F}$ to broadcast probing symbols $\mathbf{s}_{\text{probe}} \in \mathbb{C}^{N_{\mathbf{F}} \times 1}$ over the self-interference channel $\mathbf{H}$ to the receiver at the BS, which measures the broadcasted symbols using sensing beams from the columns of codebook $\mathbf{W} \in \mathbb{C}^{N_{\mathbf{r}} \times N_{\mathbf{w}}}$. The entire set of combinations of probing and sensing beams at BS receiver constitute the composite received signal $\mathbf{Y}_{\text{rx}} \in \mathbb{C}^{N_{\mathbf{w}} \times N_{\mathbf{F}}}$, which we define as

$$\mathbf{Y}_{\text{rx}} = \sqrt{P_{\text{tx}}^{\text{BS}}} \mathbf{W}^H \mathbf{H} \mathbf{F} \text{diag}(\mathbf{s}_{\text{probe}}) + \mathbf{W}^H \mathbf{N}_{\text{rx}}, \quad [\mathbf{N}_{\text{rx}}]_{i,j} \sim \mathcal{N}_{\mathbb{C}}(0, \sigma_{\text{rx}}^2), \quad (13)$$

where $\mathbf{N}_{\text{rx}} \in \mathbb{C}^{N_{\mathbf{r}} \times N_{\mathbf{F}}}$ is a matrix whose elements correspond to additive white Gaussian noise. We use the columns of $\mathbf{F}$ and $\mathbf{W}$ to construct the codebooks $\mathcal{F}$ and $\mathcal{W}$ (Fig. 1). Additionally, we assume that $\mathbf{s}_{\text{probe}} = \mathbf{1}$, and we consider the special case where $N_{\mathbf{F}} = N_{\mathbf{W}}$. We let N_{probe} denote total the number of probing beam pairs used in (13), $N_{\text{probe}} = N_{\mathbf{F}} \cdot N_{\mathbf{W}}$.

Since the full-duplex BS transmits the probing symbols $\mathbf{s}_{\text{probe}}$ using transmit and receive beams $\mathbf{f} \in \mathcal{F}$ and $\mathbf{w} \in \mathcal{W}$, we can characterize the coupled self-interference by the receive link interference-to-noise ratio (INR)

$$\text{INR}_{\text{rx}} = \frac{P_{\text{tx}}^{\text{BS}} \cdot G^2 \cdot |\mathbf{w}^H \mathbf{H} \mathbf{f}|^2}{N_t \cdot \|\mathbf{w}\|_2^2 \cdot \sigma_{\text{rx}}^2}, \quad (14)$$

where G^2 denotes the the inverse path loss derived from the isolation between the transmit and receive arrays at the BS. We consider the scenario where $\sigma_{\text{rx}}^2 = -\infty$ dB and $G^2 = 1$, leaving the analysis of how our proposed algorithm performs under varying INRs for future work.

A. Formulation of Optimization Problem

After the BS carries out downlink communications with the user equipment and receives the composite received signal $\mathbf{Y}_{\text{rx}}$, the BS utilizes (12) and (13) to jointly select the optimal narrow beam index $k_{\mathbf{V}_{\text{tx}}}^*$ and estimate the self-interference channel $\mathbf{H}$. Similar to [11], we introduce an unknown beam alignment mapping $f : \mathbf{x} \rightarrow k_{\mathbf{V}_{\text{tx}}}$ that can intelligently utilize the probing beam measurements $\mathbf{x}$ in order to predict an optimal index $k_{\mathbf{V}_{\text{tx}}}^*$ from the set $\{k_{\mathbf{V}_{\text{tx}}}\}_{i=1}^{N_{\mathbf{V}_{\text{tx}}}} := \{1, 2, \dots, N_{\mathbf{V}_{\text{tx}}}\}$. Since we wish to simultaneously compute a self-interference channel estimate $\hat{\mathbf{H}} \in \mathbb{C}^{N_{\mathbf{r}} \times N_t}$, we introduce a second mapping $g : \mathbf{Y}_{\text{rx}} \rightarrow \hat{\mathbf{H}}$ that can likewise utilize the site-

specific probing and sensing beams in an intelligent fashion. We formulate our joint optimization problem as

$$\min_{\mathbf{F}, \mathbf{W}, f, g} \gamma \mathcal{U}_{\text{DL}} + (1 - \gamma) \mathcal{U}_{\text{SI}} \quad (15a)$$

$$\text{s.t. } k_{\mathbf{V}_{\text{tx}}} = f(\mathbf{h}_{\text{tx}}^H \mathbf{F}) \quad (15b)$$

$$\hat{\mathbf{H}} = g(\mathbf{Y}_{\text{rx}}) \quad (15c)$$

$$|[\mathbf{F}]_{i,j}| = \frac{1}{\sqrt{N_{\text{t}}}}, \forall i = 1, \dots, N_{\text{t}}, \forall j = 1, \dots, N_{\mathbf{F}} \quad (15d)$$

$$|[\mathbf{W}]_{i,j}| = \frac{1}{\sqrt{N_{\text{r}}}}, \forall i = 1, \dots, N_{\text{r}}, \forall j = 1, \dots, N_{\mathbf{W}} \quad (15e)$$

$$\mathcal{U}_{\text{DL}} = \mathbb{E}_{\mathbf{h}_{\text{tx}} \in \mathcal{H}_{\text{DL}}} \left[\left| \mathbf{h}_{\text{tx}}^H [\mathbf{V}_{\text{tx}}]_{:, k_{\mathbf{V}_{\text{tx}}}} \right|^2 \right] \quad (15f)$$

$$\mathcal{U}_{\text{SI}} = \mathbb{E}_{\mathbf{H} \in \mathcal{H}_{\text{SI}}} \left[\left\| \mathbf{H} - \hat{\mathbf{H}} \right\|_{\text{F}}^2 \right], \quad (15g)$$

where $\gamma \in [0, 1]$ serves as a hyperparameter that can throttle the trade-off between optimal narrow beam selection and the estimation of $\mathbf{H}$. The term $\mathcal{H}_{\text{DL}}$ denotes the set of all possible downlink channel realizations corresponding to user equipment locations at a cell site. The second term $\mathcal{H}_{\text{SI}}$ likewise denotes the set of all possible self-interference channel realizations that manifest between the transmit and receive arrays at the BS.

B. A Multi-Task Learning NN Architecture

Similar to the optimization problem formulated in [17], our formulation in (15) is also non-convex because of the unit modulus constraints (15d) and (15e). We parameterize the probing beams $\mathbf{F}$ using a complex NN module so that the BS can learn important characteristics about the propagation environment. We equip the NN with a sensing codebook $\mathbf{W}$ containing standard DFT beams in order to ensure the sensing codebook can cover a wide enough angular space when the BS transmitter sends to the BS receiver pilot symbols designated for estimating the SI channel. We also define the narrow beam selector f and self-interference channel estimator g as separate multilayer perceptrons (MLPs). We design the NN mapping g so that it satisfies

$$g(\cdot) = g_{\text{real}}(\Re\{\cdot\}) + \text{j} \cdot g_{\text{imag}}(\Im\{\cdot\}), \quad (16)$$

where $g_{\text{real}} : \Re\{\mathbf{Y}_{\text{rx}}\} \rightarrow \Re\{\hat{\mathbf{H}}\}$ and $g_{\text{imag}} : \Im\{\mathbf{Y}_{\text{rx}}\} \rightarrow \Im\{\hat{\mathbf{H}}\}$ are separate but identical NN modules. The full NN architecture is depicted in Fig. 2.

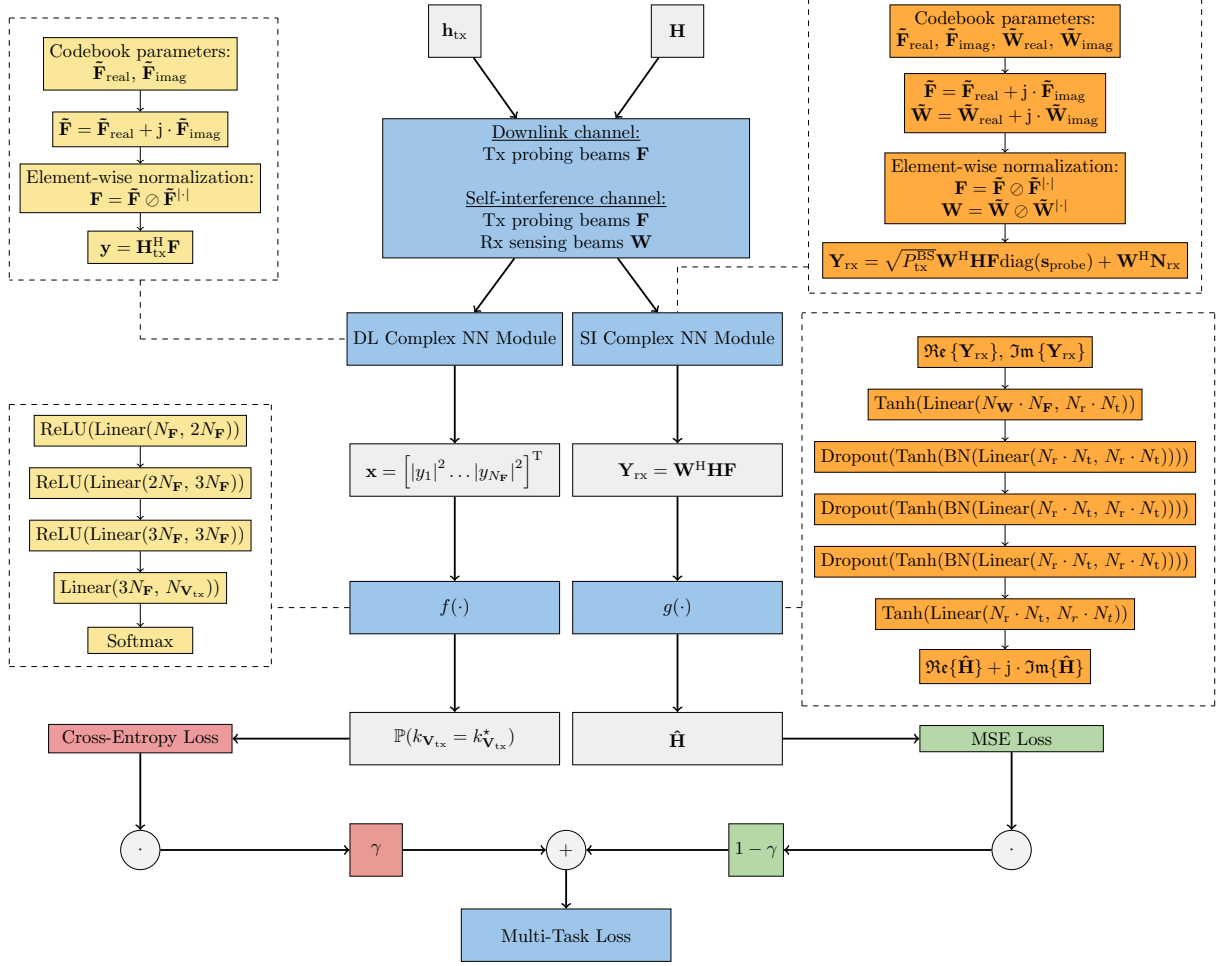


Fig. 2. The proposed NN architecture utilizes the trainable probing codebook $\mathbf{F}$ and a DFT codebook $\mathbf{W}$ to jointly optimize beam alignment and SI estimation. Given arbitrary matrices of the same dimension $\mathbf{A}$ and $\mathbf{B}$, we denote Hadamard division using $\mathbf{A} \oslash \mathbf{B}$, and we define the element-wise magnitude operator using $\mathbf{A}^{|\cdot|}$.

The beam alignment procedure over the downlink (DL) channel $\mathbf{h}_{\text{tx}}$ is adopted from [11]. In our architecture, the beam selection function f selects an optimal narrow beam index using the measurement vector $\mathbf{x}$ obtained from a sweep over $\mathcal{F}$. While $k_{\mathbf{V}_{\text{tx}}}^*$ is predicted, a self-interference complex NN module performs N_{probe} sweeps over the probing and sensing codebooks $\mathcal{F}$ and $\mathcal{W}$ in order to yield $\mathbf{Y}_{\text{rx}}$. The SI channel estimation function g then maps the composite signal to a self-interference channel estimate $\hat{\mathbf{H}}$.

C. Baselines and Metrics

Because we wish to assess whether our joint algorithm can still use the trainable, site-specific codebook $\mathbf{F}$ to achieve high beam alignment accuracy, we evaluate our method against

a version of our algorithm that largely imitates our proposed solution but instead employs a baseline DFT codebook $\mathbf{F}_{\text{DFT}} \in \mathbb{C}^{N_t \times N_F}$ to carry out a 2-tier hierarchical beam search. Rather than strategically leverage measurements about the propagation environment for narrow beam selection, this baseline version of our algorithm sweeps through N_F wide beams from the DFT probing codebook, selects the best parent wide beam, and then ultimately selects the best child beam from the narrow beam codebook $\mathbf{V}_{\text{tx}}$. The DFT codebook also replaces the trainable codebook during the calculation of the composite signal $\mathbf{Y}_{\text{rx}}$ used to estimate the self-interference coupling.

Since our joint beamforming and SI channel estimation solution seeks to estimate the self-interference coupling, we further assess our proposed site-specific algorithm against its DFT counterpart by employing the normalized mean squared error (NMSE) metric,

$$\text{NMSE} = \frac{\|\mathbf{H} - \hat{\mathbf{H}}\|_{\text{F}}^2}{\|\mathbf{H}\|_{\text{F}}^2}. \quad (17)$$

IV. DATASET

To measure the effectiveness of our proposed algorithm, we consider the line-of-sight (LOS) O1 scenario found in the DeepMIMO dataset [22]. The channel realizations between the BS and the user devices are generated using the the ray-tracing data obtained from Remcom Wireless InSite [23]. In our simulations, the BS adopts ULAs with half-wavelength spacing and operates at 28 GHz. We select BS #3 and consider 72,581 user devices located within a uniform grid that overlays the street and spans from row #800 to row #1200 in the original dataset [22], as illustrated in Fig. 3. Other DeepMIMO simulation parameters are described in Table I.

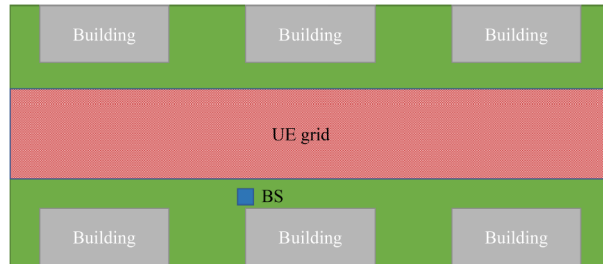


Fig. 3. Illustration of the DeepMIMO O1 scenario [17].

We leverage [24] to simulate self-interference coupling for our full duplex system. We vertically stack the transmit and receive ULAs at the BS with 20λ of separation in order to generate

72,581 realizations of the Rician SI channel. The self-interference channel simulation parameters are summarized in Table II.

TABLE I
DEEPMIMO SIMULATION PARAMETERS FOR THE BS-UE CHANNELS

Tx antenna	64×1 ULA
UE antenna	Single
Narrow beam codebook size	128
Carrier frequency	28 GHz
Bandwidth	100 MHz
Number of rays (N_p)	12

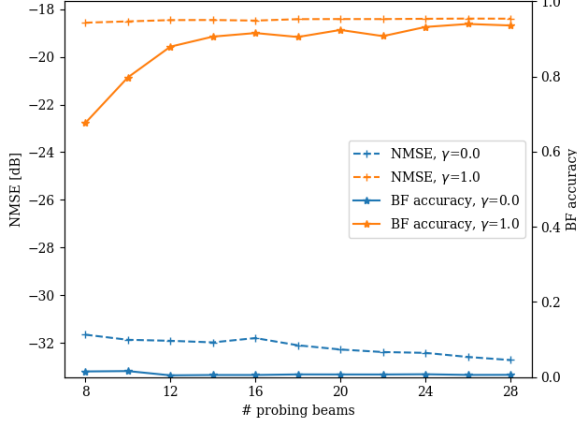
TABLE II
SIMULATION PARAMETERS FOR SELF-INTERFERENCE COUPLING

Tx antenna	64×1 ULA
Rx antenna	64×1 ULA
Carrier frequency	28 GHz
Bandwidth	100 MHz
Number of rays (M_p)	3
Rice factor (κ)	10 dB
Spacing between Tx and Rx arrays	20λ
Azimuth spread (AoA/AoD)	$[-\frac{\pi}{4}, \frac{\pi}{4}]$
Elevation spread (AoA/AoD)	$[-\frac{\pi}{2}, \frac{\pi}{2}]$

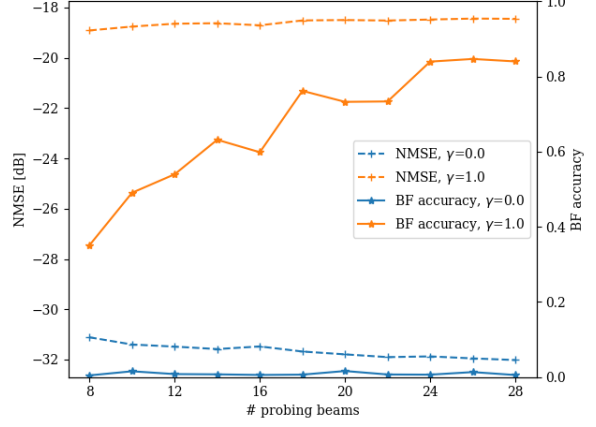
V. EVALUATION

In our experiments, we allocate 60% of downlink channel realizations to the training set, 20% to the validation set, and 20% for the test set. We apply this same train-validation-test split to our dataset of SI channel realizations. Our NNs are trained for 80 epochs with a batch size of 64 using the Adam optimizer [25]. The training process is repeated over an increasing number of probing beams N_F and over a coarse grid of hyperparameters $\gamma \in [0, 1]$.

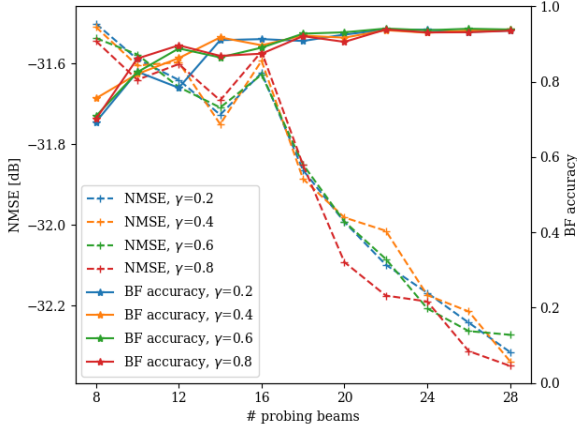
We ignore additive white Gaussian noise (AWGN) and assume that $P_{tx}^{BS} = 1$. We also separately normalize our DL and SI channel matrices by the maximum magnitude of the elements in their respective datasets: $\max_{\mathbf{h}_{tx} \in \mathcal{H}_{DL}} |[\mathbf{h}_{tx}]_{i,j}|$ and $\max_{\mathbf{H} \in \mathcal{H}_{SI}} |[\mathbf{H}]_{i,j}|$. The normalization of



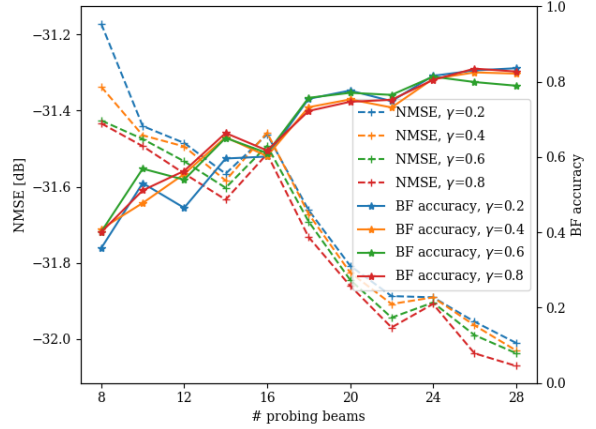
(a) Trainable codebook $\mathcal{F}$: NMSE and beam alignment accuracy vs. N_F for $\gamma \in \{0.0, 1.0\}$.



(b) DFT codebook $\mathcal{F}$: NMSE and beam alignment accuracy vs. N_F for $\gamma \in \{0.0, 1.0\}$.



(c) Trainable codebook $\mathcal{F}$: NMSE and beam alignment accuracy vs. N_F for $\gamma \in \{0.2, 0.4, 0.6, 0.8\}$.



(d) DFT codebook $\mathcal{F}$: NMSE and beam alignment accuracy vs. N_F for $\gamma \in \{0.2, 0.4, 0.6, 0.8\}$.

Fig. 4. NMSE and beam alignment accuracy vs. probing codebook size N_F for various choices of the hyperparameter γ .

the channels in $\mathcal{H}_{SI}$ is particularly critical because our NNs g_{real} and g_{imag} map their inputs to matrices whose elements fall in the $[-1, 1]$ range.

Analysis of the Multi-Task Loss. In Fig. 4 we assess the extent to which the choice of codebook $\mathcal{F}$ affects our beam alignment accuracy and NMSE metrics. A cursory comparison of Fig. 4a and Fig. 4b shows that the NN architecture in Fig. 2 (when $\gamma = 0$) attains a higher narrow beam selection accuracy when $\mathcal{F}$ is a trainable codebook rather than a DFT codebook, a result consistent with the findings of [11]. When the NN purely serves to estimate the SI channel ($\gamma = 1$), we observe that the selection of either a trainable codebook or DFT codebook does not noticeably improve or worsen estimates of the self-interference coupling.

Analysis of the Learned Probing Beams. We find that increasing the number of beams

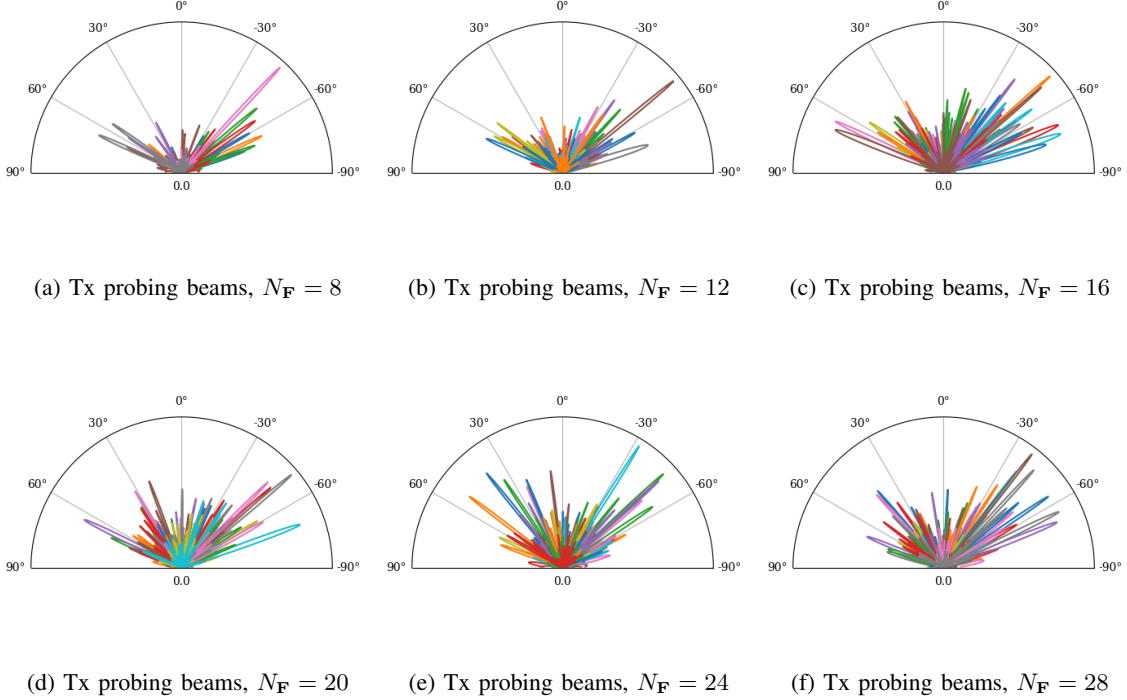


Fig. 5. Learned beam patterns ($\gamma = 0.6$) for the DeepMIMO O1 scenario under an operating frequency of 28 GHz.

N_F used by the probing and sensing codebooks simultaneously correlates with higher narrow beam alignment accuracies and lower NMSE scores for an appropriately selected γ . From Fig. 4c and Fig. 4d, we observe that the use of 28 probing beams allows the NN to not only yield highly accurate predictions of the optimal narrow beam index $k_{V_{tx}}^*$, but also jointly minimize the NMSE between $\mathbf{H}$ and $\hat{\mathbf{H}}$. We more importantly note that when the NN employs a trainable probing codebook for joint beam alignment and SI channel estimation, the trainable codebook still remains able to direct energy to capture meaningful information about the propagation environment, as illustrated by the beam patterns in Fig. 5. The loss curves in Fig. 4(c) seem to aptly reinforce this observation. We notice that as N_F increases, the NMSE metric generally decreases while the narrow beam alignment accuracy undergoes diminishing increases or even stops improving altogether. This might suggest that when the BS uses enough pilot (probing) resources during training and can no longer refine its optimal beam selection capabilities, the BS can use its remaining pilot resources to provide better estimates for the self-interference coupling.

VI. CONCLUSION

In this paper, we approach the joint beam alignment and channel estimation problem for full-duplex mmWave systems using a NN equipped with a trainable probing codebook. We empirically demonstrate that the probing codebook at the BS can capture site-specific characteristics that can be used to predict an optimal beam, even while the NN simultaneously refines its estimates of the self-interference channel. Our experiments suggest that employing a trainable probing codebook rather than a DFT probing codebook does not adversely interfere with the estimation of the SI channel and can facilitate better beam alignment accuracies. Future works could assess whether our joint beam alignment and channel estimation algorithm still performs well in environments characterized by many non-line-of-sight (NLOS) users. Ultimately, our study motivates a variety of future works related to self-interference estimation and cancellation during the beam alignment process.

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